

The screenshot displays the Visual Studio Code (VS Code) interface with a Java project named "java lab sheet 4".

Explorer (Left Sidebar): Shows the project structure with files Q11.java through Q26.java. The "java lab sheet 4" folder is expanded, showing all these files.

Main Editor: Displays the code for Q11.java. The code defines a `Calculator` class with the following methods:

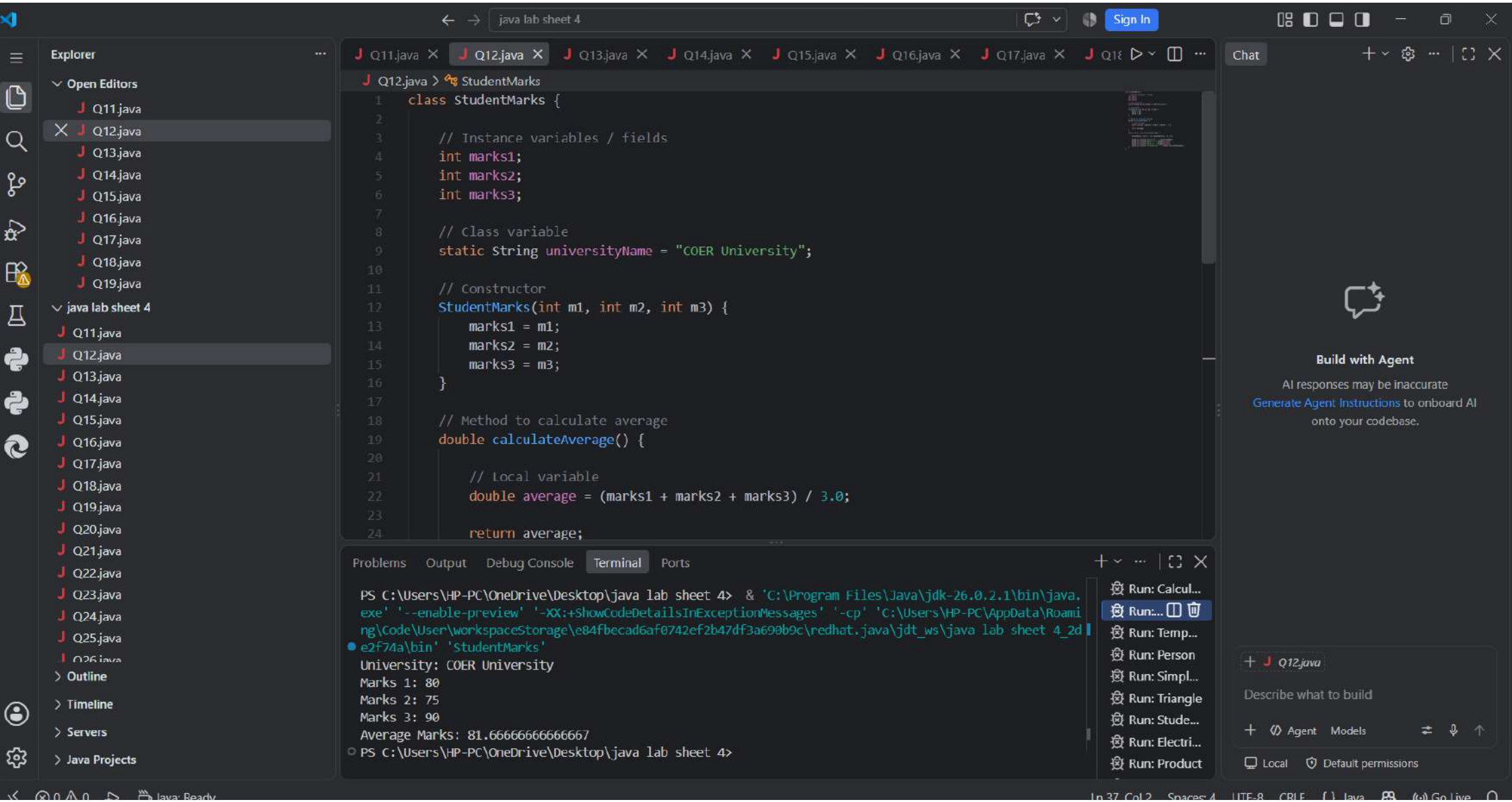
- `add(int a, int b)`: Returns the sum of `a` and `b`.
- `subtract(int a, int b)`: Returns the difference of `a` and `b`.
- `multiply(int a, int b)`: Returns the product of `a` and `b`.
- `divide(int a, int b)`: Returns the quotient of `a` and `b`.
- `main(String[] args)`: The entry point of the program, which creates a `Calculator` object and performs the following operations:
 - Addition: `20 + 10 = 30`
 - Subtraction: `20 - 10 = 10`
 - Multiplication: `20 * 10 = 200`
 - Division: `20 / 10 = 2.0`
 - Total Operations: 4

Output Window (Bottom): Shows the execution results of the program:

```
PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '-enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2de2f74a\bin' 'calculator'
Addition: 30
Subtraction: 10
Multiplication: 200
Division: 2.0
Total Operations: 4
PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>
```

Run and Debug (Bottom Right): Shows the "Run" button and a list of run configurations: Run: Stude..., Run: Temp..., Run: Person, Run: Simpl..., Run: Triangle, Run: Stude..., Run: Electri..., Run: Product.

Status Bar (Bottom): Shows the current cursor position: Line 45, Column 2. The language is set to Java.



Sign In

java lab sheet 4

Q11.java

Q12.java

Q13.java

Q14.java

Q15.java

Q16.java

Q17.java

Q18.java

Q19.java

Chat

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Explorer

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Q15.java

Q16.java

Q17.java

Q18.java

Q19.java

java lab sheet 4

Q11.java

Q12.java

Q13.java

Q14.java

Q15.java

Q16.java

Q17.java

Q18.java

Q19.java

Q20.java

Q21.java

Q22.java

Q23.java

Q24.java

Q25.java

Q26.java

Outline

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Java Projects

Q14.java

Person

1 class Person {

2

3 String name;

4 int age;

5

6 static String country = "India";

7

8 Person(String name, int age) {

9 this.name = name;

10 this.age = age;

11 }

12

13 void checkVotingEligibility() {

14

15 boolean eligible = age >= 18;

16

17 System.out.println("Name: " + name);

18 System.out.println("Age: " + age);

19 System.out.println("Country: " + country);

20

21 if (eligible) {

22 System.out.println(x: "Eligible for voting");

23 } else {

24 System.out.println(x: "Not eligible for voting");

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'c:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2d\bin' 'Person'

Name: Bhavna

Age: 19

Country: India

Eligible for voting

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>

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...

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Run: Calcul...

Run: Stude...

Run: Temp...

Run: ...

Run: Simpl...

Run: Triangle

Run: Stude...

Run: Electri...

Run: Product

+ Q14.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 33, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

☰

Explorer

▼ Open Editors

J Q11.java

J Q12.java

J Q13.java

J Q14.java

X J Q15.java

J Q16.java

J Q17.java

J Q18.java

J Q19.java

▼ java lab sheet 4

J Q11.java

J Q12.java

J Q13.java

J Q14.java

J Q15.java

J Q16.java

J Q17.java

J Q18.java

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J Q20.java

J Q21.java

J Q22.java

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J Q25.java

J Q26.java

Outline

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← → java lab sheet 4

Sign In

J Q11.java X J Q12.java X J Q13.java X J Q14.java X J Q15.java X J Q16.java X J Q17.java X J Q18.java X

J Q15.java > SimpleInterest

```
1 class SimpleInterest {
2
3     double principal;
4     double rate;
5     double time;
6
7     static String bank = "State Bank";
8
9     SimpleInterest(double principal, double rate, double time) {
10         this.principal = principal;
11         this.rate = rate;
12         this.time = time;
13     }
14
15     void calculateInterest() {
16
17         double interest = (principal * rate * time) / 100;
18         double totalAmount = principal + interest;
19
20         System.out.println("Bank: " + bank);
21         System.out.println("Principal: " + principal);
22         System.out.println("Rate: " + rate + "%");
23         System.out.println("Time: " + time + " years");
24         System.out.println("Simple Interest: " + interest);
25     }
26 }
```

Problems Output Debug Console Terminal Ports

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2d...e2f74a\bin' 'SimpleInterest'

Bank: State Bank
Principal: 10000.0
Rate: 5.0%
Time: 2.0 years
Simple Interest: 1000.0
Total Amount: 11000.0

Run: Calcul...
Run: Stude...
Run: Temp...
Run: Person
Run: ...
Run: Triangle
Run: Stude...
Run: Electri...
Run: Product

Chat

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Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

✕ 0 0 0 0 Java: Ready

Ln 33, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

Sign in

Chat

Q11.java Q12.java Q13.java Q14.java Q15.java Q16.java Q17.java Q18.java Q19.java

Q11.java Q12.java Q13.java Q14.java Q15.java Q16.java Q17.java Q18.java Q19.java Q20.java Q21.java Q22.java Q23.java Q24.java Q25.java Q26.java

Outline Timeline Servers Java Projects

SimpleInterest

1 class SimpleInterest {
15 void calculateInterest() {
16
17 double interest = (principal * rate * time) / 100;
18 double totalAmount = principal + interest;
19
20 System.out.println("Bank: " + bank);
21 System.out.println("Principal: " + principal);
22 System.out.println("Rate: " + rate + "%");
23 System.out.println("Time: " + time + " years");
24 System.out.println("Simple Interest: " + interest);
25 System.out.println("Total Amount: " + totalAmount);
26 }
27
28 Run | Debug
29 public static void main(String[] args) {
30
31 SimpleInterest s = new SimpleInterest(principal: 10000, rate: 5, time: 2);
32 s.calculateInterest();
33 }
34

Problems Output Debug Console Terminal Ports

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a590b9c\redhat.java\jdt_ws\java lab sheet 4_2de2f74a\bin' 'SimpleInterest'
Principal: 10000.0
Rate: 5.0%
Time: 2.0 years
Simple Interest: 1000.0
Total Amount: 11000.0
PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>

Run: Calcul...
Run: Stude...
Run: Temp...
Run: Person...
Run: Run...
Run: Triangle...
Run: Stude...
Run: Electri...
Run: Product...

Build with Agent

AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.

Q15.java

Describe what to build

Agent Models

Local Default permissions

Ln 33, Col 2 Spaces: 4 UTF-8 CR LF {} Java Go Live

The image shows a screenshot of the Visual Studio Code (VS Code) editor interface. The main editor window displays a Java file named Q16.java, which contains a class named Triangle. The class has three integer attributes (a, b, c) and a static String attribute (type) representing the triangle type. The Triangle constructor initializes the attributes and sets the type based on the side lengths. The checkTriangle method checks if the triangle is valid (satisfies the triangle inequality) and returns a boolean. The toString method returns the triangle type. The terminal window at the bottom shows the command to run the program, and the output indicates that the triangle is valid and its type is Isosceles. The Explorer sidebar on the left shows the file structure, and the Chat sidebar on the right is visible.

Explorer

Open Editors

Q11.java

Q12.java

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Q14.java

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Q19.java

java lab sheet 4

Q11.java

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java lab sheet 4

Q11.java Q12.java Q13.java Q14.java Q15.java Q16.java Q17.java Q18.java

Q16.java

Triangle

1class Triangle {

15void checkTriangle() {

23if (a == b && b == c) {

24triangleType = "Equilateral";

25} else if (a == b || b == c || a == c) {

26triangleType = "Isosceles";

27} else {

28triangleType = "Scalene";

29}

30

31System.out.println(x: "Triangle is valid.");

32System.out.println("Triangle Type: " + triangleType);

33

34} else {

35System.out.println(x: "Triangle is not valid.");

36}

37}

38

39public static void main(String[] args) {

40

41Triangle t = new Triangle(a: 5, b: 5, c: 6);

42t.checkTriangle();

43

Run | Debug

public static void main(String[] args) {

40

41Triangle t = new Triangle(a: 5, b: 5, c: 6);

42t.checkTriangle();

43

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2d\bin' 'Triangle'

Triangle is valid.

Triangle Type: Isosceles

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>

Run: Calcul...

Run: Stude...

Run: Temp...

Run: Person

Run: Simpl...

Run: ...

Run: Stude...

Run: Electri...

Run: Product

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ Q16.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 1, Col 17 Spaces: 4 UTF-8 CRLF {} Java Go Live

☰

Explorer

▼ Open Editors

Q11.java

Q12.java

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✕ Q16.java

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▼ java lab sheet 4

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Q11.java ✕

Q12.java ✕

Q13.java ✕

Q14.java ✕

Q15.java ✕

Q16.java ✕

Q17.java ✕

Q18.java ▶

Q19.java ▶

Q20.java ▶

Q21.java ▶

Q22.java ▶

Q23.java ▶

Q24.java ▶

Q25.java ▶

Q26.java ▶

Q16.java > Triangle

1 class Triangle {

15 void checkTriangle() {

33

34 } else {

35 System.out.println(x: "Triangle is not valid.");

36 }

37 }

38

Run | Debug

39 public static void main(String[] args) {

40

41 Triangle t = new Triangle(a: 5, b: 5, c: 6);

42 t.checkTriangle();

43 }

44 }

45

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '-enable-preview' '-XX:showCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2d\bin' 'Triangle'

Triangle is valid.

Triangle Type: Isosceles

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>

Run: Calcul...

Run: Stude...

Run: Temp...

Run: Person

Run: Simpl...

Run: ...

Run: Stude...

Run: Electri...

Run: Product

Chat

+ ✕ ⚙ ... [] ✕

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ Q16.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 1, Col 17 Spaces: 4 UTF-8 CRLF {} Java Go Live

☰

Explorer

▼ Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Q15.java

Q16.java

✕ Q17.java

Q18.java

Q19.java

▼ java lab sheet 4

Q11.java

Q12.java

Q13.java

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Outline

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← →

java lab sheet 4

Sign In

Q11.java ✕

Q12.java ✕

Q13.java ✕

Q14.java ✕

Q15.java ✕

Q16.java ✕

Q17.java ✕

Q18.java ▶

Q19.java ▶

...

Q17.java > StudentGrade

```
1  class StudentGrade {
13     void assignGrade() {
23         } else if (marks >= 70) {
24             grade = "B";
25         } else if (marks >= 60) {
26             grade = "C";
27         } else {
28             grade = "D";
29         }
30
31         System.out.println("Name: " + name);
32         System.out.println("Marks: " + marks);
33         System.out.println("Grade: " + grade);
34     }
35
36     Run | Debug
37     public static void main(String[] args) {
38
39         StudentGrade s = new StudentGrade(name: "Bhavna", marks: 85);
40         s.assignGrade();
41     }
42 }
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-25.0.2.1\bin\java.
exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roami
ng\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2d
e2f74a\bin' 'StudentGrade'
Name: Bhavna
Marks: 85
Grade: A
PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>
```

Run: Calcul...

Run: Stude...

Run: Temp...

Run: Person

Run: Simpl...

Run: Triangle

Run: Run... [icon] [icon]

Run: Electri...

Run: Product

Chat

+ ✕ ⚙ ... [icon] [icon]

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ Q17.java

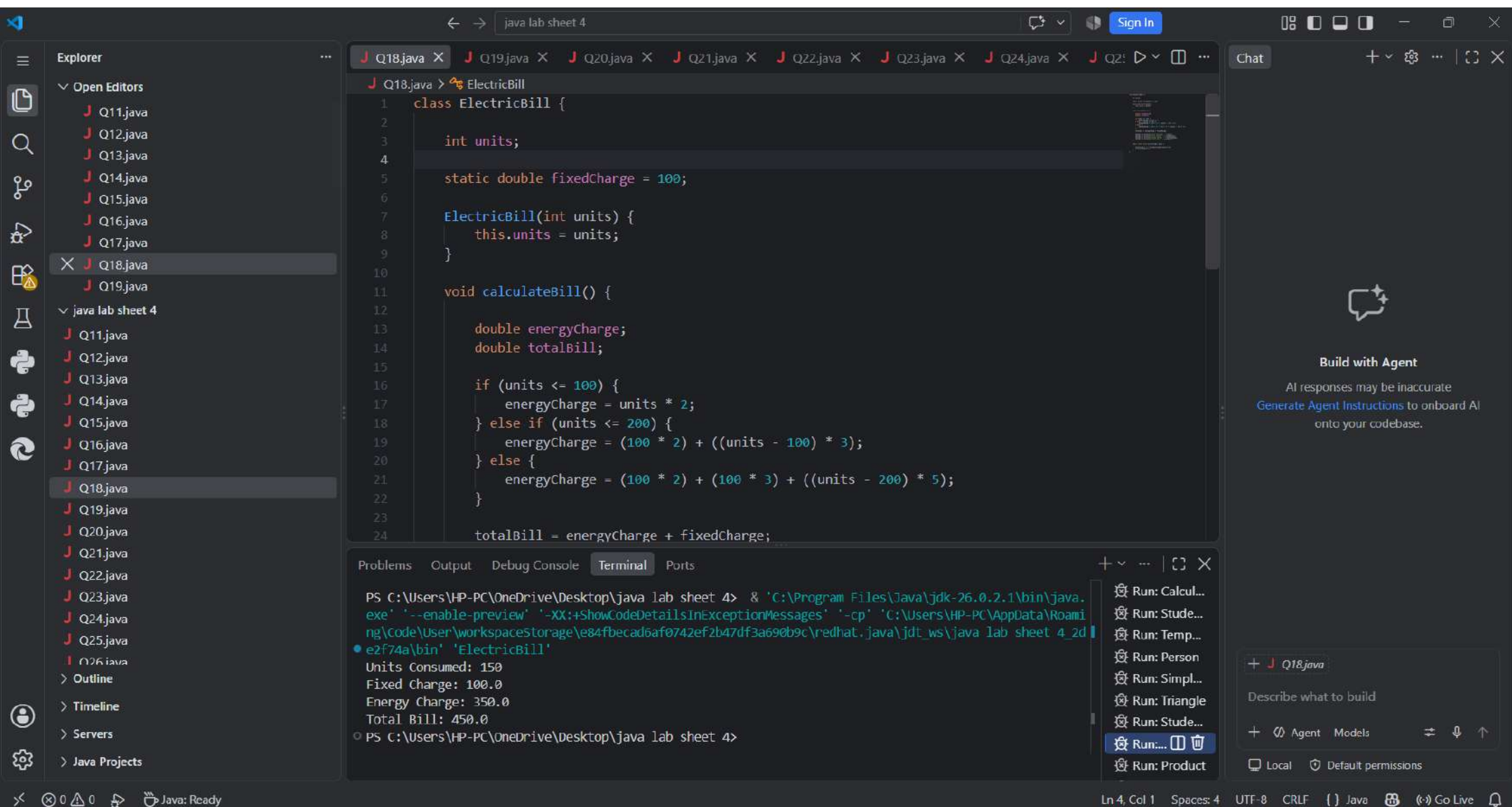
Describe what to build

+ [icon] Agent Models [icon] [icon] [icon]

Local [icon] Default permissions

< 0 0 0 Java: Ready

Ln 41, Col 2 Spaces: 4 UTF-8 CRLF {} Java [icon] Go Live [icon]



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Explorer

▼ Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Q15.java

Q16.java

Q17.java

Q18.java

Q19.java

▼ java lab sheet 4

Q11.java

Q12.java

Q13.java

Q14.java

Q15.java

Q16.java

Q17.java

Q18.java

Q19.java

Q20.java

Q21.java

Q22.java

Q23.java

Q24.java

Q25.java

Q26.java

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← → java lab sheet 4

Q18.java Q19.java Q20.java Q21.java Q22.java Q23.java Q24.java Q25.java

Q19.java > Product

```
1
2 class Product {
3
4     int id;
5     String name;
6     double price;
7
8     static double discountRate = 10;
9
10    Product(int id, String name, double price) {
11        this.id = id;
12        this.name = name;
13        this.price = price;
14    }
15
16    void calculateFinalPrice() {
17
18        double discount = (price * discountRate) / 100;
19        double finalPrice = price - discount;
20
21        System.out.println("Product ID: " + id);
22        System.out.println("Product Name: " + name);
23        System.out.println("Original Price: " + price);
24        System.out.println("Discount: " + discount);
25    }
26}
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '-enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\e84fbecad6af0742ef2b47df3a690b9c\redhat.java\jdt_ws\java lab sheet 4_2d\bin' 'Product'

Product ID: 101

Product Name: Laptop

Original Price: 50000.0

Discount: 5000.0

Final Price: 45000.0

PS C:\Users\HP-PC\OneDrive\Desktop\java lab sheet 4>

Run... ⌘ W

Run: BankA...

Run: Palind...

Run: Armst...

Run: Prime...

Run: Fibon...

Run: Factor...

Run: Matrix

Run: String...

Run: Word...

Chat

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Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ Q19.java

Describe what to build

+ ⚙ Agent Models ⚙ ⬇ ⬆

Local Default permissions

Ln 33, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

